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Method and system for switching a dispensed fluid in a nanofabrication dispensing system

Abstract

A method of switching a first fluid supplied to a dispenser with a second fluid. The method includes stopping the first fluid from flowing into a common portion of a flow system, causing a flush fluid to flow through the common portion, stopping the flush fluid from flowing into the common portion, removing the flush fluid from the common portion, causing the second fluid to flow through the common portion, and continuously circulating the first fluid to and from a reservoir without the first liquid entering the common portion of the flow system. The continuously circulating of the first fluid begins a) after stopping the fluid liquid from flowing into the common portion of the flow system and b) prior to or during the second fluid flowing through the common portion of the flow system.

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Background/Summary

BACKGROUND

Field of Art

(1) The present disclosure relates to a fluid switching method in a dispensing system, and in particular, a method of switching a first fluid supplied to a dispenser with a second fluid in a nanofabrication system.

Description of the Related Art

(2) Nanofabrication includes the fabrication of very small structures that have features on the order of 100 nanometers or smaller. One application in which nano-fabrication has had a sizeable impact is in the fabrication of integrated circuits. The semiconductor processing industry continues to

strive for larger production yields while increasing the circuits per unit area formed on a substrate. Improvements in nano-fabrication include providing greater process control and/or improving throughput while also allowing continued reduction of the minimum feature dimensions of the structures formed.

(3) Nanofabrication techniques include nanoimprinting and planarizing. Nanoimprint and planarization is useful in a variety of applications including, for example, fabricating one or more layers of integrated devices by shaping a film on a substrate. Examples of an integrated device include but are not limited to CMOS logic, microprocessors, NAND Flash memory, NOR Flash memory, DRAM memory, MRAM, 3D cross-point memory, Re-RAM, Fe-RAM, STT-RAM, MEMS, optical components, and the like. Exemplary nanofabrication systems using nanoimprinting and planarizing processes are described in detail in numerous publications, such as U.S. Pat. Nos. 8,349,241, 8,066,930, and 6,936,194.

(4) The technique disclosed in each of the aforementioned patents describes the shaping of a film on a substrate by the formation of a relief pattern in a formable material (polymerizable) layer. The shape of this film may then be used to transfer a pattern corresponding to the relief pattern into and/or onto an underlying substrate.

(5) The patterning process uses a template spaced apart from the substrate and the formable material is applied between the template and the substrate. The template is brought into contact with the formable material causing the formable material to spread and fill the space between the template and the substrate. The formable liquid is solidified to form a film that has a shape (pattern) conforming to a shape of the surface of the template that is in contact with the formable liquid. After solidification, the template is separated from the solidified layer such that the template and the substrate are spaced apart.

(6) A planarization technique sometimes referred to as inkjet-based adaptive planarization (IAP) involves dispensing a variable drop pattern of formable material between the substrate and a superstrate, where the drop pattern varies depending on the substrate topography. A superstrate is then brought into contact with the formable material after which the material is polymerized on the substrate, and the superstrate removed.

(7) The substrate and the solidified layer may then be subjected to additional processes, such as etching processes, to transfer an image into the substrate that corresponds to the pattern in one or both of the solidified layer and/or patterned layers that are underneath the solidified layer. The patterned substrate can be further subjected to known steps and processes for device (article) fabrication, including, for example, curing, oxidation, layer formation, deposition, doping, planarization, etching, formable material removal, dicing, bonding, and packaging, and the like.

(8) In nanofabrication systems, depending on the product being manufactured, the process requires switching from dispensing one formable material to dispensing another formable material. In known systems, switching materials can take significant time and waste a great deal of formable material, which increases costs. Typically, in conventional systems, in order to change the formable material supplied to the dispenser, the operator must shut down and drain the existing first material from a reservoir, fill the reservoir with a second material, circulate the second material, shut down and drain the second material, fill the reservoir again with more of the second material, circulate the new second material again, and repeat the steps of filling/circulating/draining two or more times. Thus, there is a need in the art for a method of switching a first fluid supplied to a dispenser with a second fluid that takes less time and avoids wasting materials.

SUMMARY

(9) A method of switching a first fluid supplied to a dispenser with a second fluid, the method comprises stopping the first fluid from flowing into a common portion of a flow system, the common portion including: a supply manifold, the dispenser, a return manifold, a supply path providing fluid communication between the supply manifold and the dispenser, and a return path providing fluid communication between the dispenser and the return manifold, causing a flush fluid

to flow through the common portion of the flow system, stopping the flush fluid from flowing into the common portion of the flow system, removing the flush fluid from the common portion of the flow system, causing the second fluid to flow through the common portion of the flow system, and continuously circulating the first fluid to and from a reservoir without the first liquid entering the common portion of the flow system, wherein the continuously circulating of the first fluid begins a) after stopping the fluid liquid from flowing into the common portion of the flow system and b) prior to or during the second fluid flowing through the common portion of the flow system.

(10) A dispensing system comprises a flow system having a common portion, the common portion including a supply manifold, a dispenser, a return manifold, a supply path providing fluid communication between the supply manifold and the dispenser, and a return path providing fluid communication between the dispenser and the return manifold, one or more processors, and one or more memories storing instructions, when executed by the one or more processors, causes the dispensing system to: stop a first fluid from flowing into the common portion of a flow system, cause a flush fluid to flow through the common portion of the flow system, stop the flush fluid from flowing into the common portion of the flow system, remove the flush fluid from the common portion of the flow system, cause a second fluid to flow through the common portion of the flow system, and continuously circulate the first fluid to and from a reservoir without the first liquid entering the common portion of the flow system, wherein the continuously circulating of the first fluid begins a) after stopping the fluid liquid from flowing into the common portion of the flow system and b) prior to or during the second fluid flowing through the common portion of the flow system.

(11) A method of making an article comprises stopping a first fluid from flowing into a common portion of a flow system, the common portion including a supply manifold, the dispenser, a return manifold, a supply path providing fluid communication between the supply manifold and the dispenser, and a return path providing fluid communication between the dispenser and the return manifold, causing a flush fluid to flow through the common portion of the flow system, stopping the flush fluid from flowing into the common portion of the flow system, removing the flush fluid from the common portion of the flow system, causing the second fluid to flow through the common portion of the flow system, continuously circulating the first fluid to and from a reservoir without the first liquid entering the common portion of the flow system, wherein the continuously circulating of the first fluid begins a) after stopping the fluid liquid from flowing into the common portion of the flow system and b) prior to or during the second fluid flowing through the common portion of the flow system, dispensing the second fluid onto a substrate using the dispenser, forming a pattern or a layer of the dispensed fluid on the substrate, and processing the formed pattern or layer to make the article.

(12) These and other objects, features, and advantages of the present disclosure will become apparent upon reading the following detailed description of exemplary embodiments of the present disclosure, when taken in conjunction with the appended drawings, and provided claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) So that features and advantages of the present disclosure can be understood in detail, a more particular description of embodiments of the disclosure may be had by reference to the embodiments illustrated in the appended drawings. It is to be noted, however, that the appended drawings only illustrate typical embodiments of the disclosure, and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is an illustration of an example nanofabrication system in accordance with an example

embodiment.

(3) FIG. 2 is an illustration of an example template in accordance with an example embodiment.

(4) FIG. 3 is a flowchart illustrating an example nanofabrication method in accordance with an example embodiment.

(5) FIG. 4 is a schematic representation of a flow system in accordance with an example embodiment.

(6) FIG. 5 is a schematic representation of material management system in accordance with an example embodiment.

(7) FIG. 6 is a schematic representation of a flush system in accordance with an example embodiment.

(8) FIG. 7 is a flow chart of a method of switching a first fluid supplied to a dispenser with a second fluid in accordance with an example embodiment.

(9) FIG. 8 is a schematic representation of the flow system of FIG. 4 in a state when the first fluid is flowing into a common portion of the flow system and with all other fluids in a recycle bypass.

(10) FIG. 9 is a schematic representation of the flow system of FIG. 4 in a state when the first fluid is stopped from flowing and a flush fluid is stopped from flowing, with a third and fourth fluid in a recycle bypass.

(11) FIG. 10 is a schematic representation of the flow system of FIG. 4 in a state when a flush fluid is flowing through the common portion of the flow system with all other fluids in a recycle bypass.

(12) FIG. 11 is a schematic representation of the flow system of FIG. 4 in a state when the flush fluid is stopped from flowing and a second fluid is stopped from flowing, with the first fluid and the third fluid in a recycle bypass.

(13) FIG. 12 is a schematic representation of the flow system of FIG. 4 in a state when the second fluid is flowing into the common portion of the flow system, with all other fluids in a recycle bypass.

(14) Throughout the figures, the same reference numerals and characters, unless otherwise stated, are used to denote like features, elements, components or portions of the illustrated embodiments. Moreover, while the subject disclosure will now be described in detail with reference to the figures, it is done so in connection with the illustrative exemplary embodiments. It is intended that changes and modifications can be made to the described exemplary embodiments without departing from the true scope and spirit of the subject disclosure as defined by the appended claims.

DETAILED DESCRIPTION

(15) Throughout this disclosure, reference is made primarily to nanoimprint lithography, which uses the above-mentioned patterned template to impart a pattern onto formable liquid. However, as mentioned below, in an alternative embodiment, the template is featureless in which case a planar surface may be formed on the substrate. In such embodiments where a planar surface is formed, the formation process is referred to as planarization. Thus, throughout this disclosure, whenever nanoimprint lithography is mentioned, it should be understood that the same method is applicable to planarization. The term superstrate is used in place of the term template in instances where the template is featureless.

(16) As noted above, a known method of switching fluids to a dispenser in fabrication requires significant time and causes material waste. Thus, it is desirable for a method of switching a first fluid supplied to a dispenser with a second fluid that avoids these disadvantages.

(17) Nanofabrication System (Shaping System)

(18) FIG. 1 is an illustration of a nanoimprint lithography system **100** in which an embodiment may be implemented. The nanoimprint lithography system **100** is used to shape a film on a substrate **102**. The substrate **102** may be coupled to a substrate chuck **104**. The substrate chuck **104** may be but is not limited to a vacuum chuck, pin-type chuck, groove-type chuck, electrostatic chuck, electromagnetic chuck, and/or the like.

(19) The substrate **102** and the substrate chuck **104** may be further supported by a substrate

positioning stage **106**. The substrate positioning stage **106** may provide translational and/or rotational motion along one or more of the x, y, z, θ , and ϕ -axes. The substrate positioning stage **106**, the substrate **102**, and the substrate chuck **104** may also be positioned on a base (not shown). The substrate positioning stage may be a part of a positioning system.

(20) Spaced-apart from the substrate **102** is a template **108**. The template **108** may include a body having a mesa (also referred to as a mold) **110** extending towards the substrate **102** on a front side of the template **108**. The mesa **110** may have a patterning surface **112** thereon also on the front side of the template **108**. Alternatively, the template **108** may be formed without the mesa **110**, in which case the surface of the template facing the substrate **102** is equivalent to the mold **110** and the patterning surface **112** is that surface of the template **108** facing the substrate **102**.

(21) The template **108** may be formed from such materials including, but not limited to, fused-silica, quartz, silicon, organic polymers, siloxane polymers, borosilicate glass, fluorocarbon polymers, metal, hardened sapphire, and/or the like. The patterning surface **112** may have features defined by a plurality of spaced-apart template recesses **114** and/or template protrusions **116**. The patterning surface **112** defines a pattern that forms the basis of a pattern to be formed on the substrate **102**. In an alternative embodiment, the patterning surface **112** is featureless in which case a planar surface is formed on the substrate. In an alternative embodiment, the patterning surface **112** is featureless and the same size as the substrate and a planar surface is formed across the entire substrate. In such embodiments where a planar surface is formed, the formation process may be alternatively referred to as planarization and the featureless template may be alternatively referred to as a superstrate.

(22) The template **108** may be coupled to a template chuck **118**. The template chuck **118** may be, but is not limited to, vacuum chuck, pin-type chuck, groove-type chuck, electrostatic chuck, electromagnetic chuck, and/or other similar chuck types. The template chuck **118** may be configured to apply stress, pressure, and/or strain to template **108** that varies across the template **108**. The template chuck **118** may include piezoelectric actuators which can squeeze and/or stretch different portions of the template **108**. The template chuck **118** may include a system such as a zone based vacuum chuck, an actuator array, a pressure bladder, etc. which can apply a pressure differential to a back surface of the template causing the template to bend and deform.

(23) The template chuck **118** may be coupled to an imprint head **120** which is a part of the positioning system. The imprint head may be moveably coupled to a bridge. The imprint head may include one or more actuators such as voice coil motors, piezoelectric motors, linear motor, nut and screw motor, etc., which are configured to move the template chuck **118** relative to the substrate in at least the z-axis direction, and potentially other directions (e.g., x, y, θ , ψ , and ϕ -axes).

(24) The nanoimprint lithography system **100** may further comprise a fluid supply and dispensing system **200** that includes a fluid dispenser **122** among other components discussed below in more detail. The fluid dispenser **122** may also be moveably coupled to the bridge. In an embodiment, the fluid dispenser **122** and the imprint head **120** share one or more or all positioning components. In an alternative embodiment, the fluid dispenser **122** and the imprint head **120** move independently from each other. The fluid dispenser **122** may be used to deposit liquid formable material **124** (e.g., polymerizable material) onto the substrate **102** in a pattern. Additional formable material **124** may also be added to the substrate **102** using techniques, such as, drop dispense, spin-coating, dip coating, chemical vapor deposition (CVD), physical vapor deposition (PVD), thin film deposition, thick film deposition, and/or the like prior to the formable material **124** being deposited onto the substrate **102**. The formable material **124** may be dispensed upon the substrate **102** before and/or after a desired volume is defined between the mold **112** and the substrate **102** depending on design considerations. The formable material **124** may comprise a mixture including a monomer as described in U.S. Pat. Nos. 7,157,036 and 8,076,386, both of which are herein incorporated by reference. The formable material is alternatively referred herein simply as “fluid” or “fluid material.” The formable material (fluid, fluid material) is in a liquid state. Furthermore, the terms

“first fluid,” “second fluid,” etc. refers to different instances of the formable material

(25) Different fluid dispensers **122** may use different technologies to dispense formable material **124**. When the formable material **124** is jettable, ink jet type dispensers may be used to dispense the formable material. For example, thermal ink jetting, microelectromechanical systems (MEMS) based ink jetting, valve jet, and piezoelectric ink jetting are common techniques for dispensing jettable liquids.

(26) The nanoimprint lithography system **100** may further comprise a radiation source **126** that directs actinic energy along an exposure path **128**. The imprint head and the substrate positioning stage **106** may be configured to position the template **108** and the substrate **102** in superimposition with the exposure path **128**. The radiation source **126** sends the actinic energy along the exposure path **128** after the template **108** has made contact with the formable material **128**. FIG. **1** illustrates the exposure path **128** when the template **108** is not in contact with the formable material **124**, this is done for illustrative purposes so that the relative position of the individual components can be easily identified. An individual skilled in the art would understand that exposure path **128** would not substantially change when the template **108** is brought into contact with the formable material **124**.

(27) The nanoimprint lithography system **100** may further comprise a field camera **136** that is positioned to view the spread of formable material **124** after the template **108** has made contact with the formable material **124**. FIG. **1** illustrates an optical axis of the field camera's imaging field as a dashed line. As illustrated in FIG. **1** the nanoimprint lithography system **100** may include one or more optical components (dichroic mirrors, beam combiners, prisms, lenses, mirrors, etc.) which combine the actinic radiation with light to be detected by the field camera. The field camera **136** may be configured to detect the spread of formable material under the template **108**. The optical axis of the field camera **136** as illustrated in FIG. **1** is straight but may be bent by one or more optical components. The field camera **136** may include one or more of a CCD, a sensor array, a line camera, and a photodetector which are configured to gather light that has a wavelength that shows a contrast between regions underneath the template **108** that are in contact with the formable material, and regions underneath the template **108** which are not in contact with the formable material **124**. The field camera **136** may be configured to gather monochromatic images of visible light. The field camera **136** may be configured to provide images of the spread of formable material **124** underneath the template **108**, the separation of the template **108** from cured formable material, and can be used to keep track the progress over the imprinting process.

(28) The nanoimprint lithography system **100** may further comprise a droplet inspection system **138** that is separate from the field camera **136**. The droplet inspection system **138** may include one or more of a CCD, a camera, a line camera, and a photodetector. The droplet inspection system **138** may include one or more optical components such as lenses, mirrors, apertures, filters, prisms, polarizers, windows, adaptive optics, and/or light sources. The droplet inspection system **138** may be positioned to inspect droplets prior to the patterning surface **112** contacting the formable material **124** on the substrate **102**.

(29) The nanoimprint lithography system **100** may further include a thermal radiation source **134** which may be configured to provide a spatial distribution of thermal radiation to one or both of the template **108** and the substrate **102**. The thermal radiation source **134** may include one or more sources of thermal electromagnetic radiation that will heat up one or both of the substrate **102** and the template **108** and does not cause the formable material **124** to solidify. The thermal radiation source **134** may include a spatial light modulator such as a digital micromirror device (DMD), Liquid Crystal on Silicon (LCoS), Liquid Crystal Device (LCD), etc., to modulate the spatial temporal distribution of thermal radiation. The nanoimprint lithography system may further comprise one or more optical components which are used to combine the actinic radiation, the thermal radiation, and the radiation gathered by the field camera **136** onto a single optical path that intersects with the imprint field when the template **108** comes into contact with the formable

material **124** on the substrate **102**. The thermal radiation source **134** may send the thermal radiation along a thermal radiation path (which in FIG. **1** is illustrated as 2 thick dark lines) after the template **108** has made contact with the formable material **124**. FIG. **1** illustrates the thermal radiation path when the template **108** is not in contact with the formable material **124**, this is done for illustrative purposes so that the relative position of the individual components can be easily identified. An individual skilled in the art would understand that the thermal radiation path would not substantially change when the template **108** is brought into contact with the formable material **124**. In FIG. **1** the thermal radiation path is shown terminating at the template **108**, but it may also terminate at the substrate **102**. In an alternative embodiment, the thermal radiation source **134** is underneath the substrate **102**, and thermal radiation path is not combined with the actinic radiation and visible light.

(30) Prior to the formable material **124** being dispensed onto the substrate, a substrate coating **132** may be applied to the substrate **102**. In an embodiment, the substrate coating **132** may be an adhesion layer. In an embodiment, the substrate coating **132** may be applied to the substrate **102** prior to the substrate being loaded onto the substrate chuck **104**. In an alternative embodiment, the substrate coating **132** may be applied to substrate **102** while the substrate **102** is on the substrate chuck **104**. In an embodiment, the substrate coating **132** may be applied by spin coating, dip coating, etc. In an embodiment, the substrate **102** may be a semiconductor wafer. In another embodiment, the substrate **102** may be a blank template (replica blank) that may be used to create a daughter template after being imprinted.

(31) The nanoimprint lithography system **100** may be regulated, controlled, and/or directed by one or more processors **140** (controller) in communication with one or more components and/or subsystems such as the substrate chuck **104**, the substrate positioning stage **106**, the template chuck **118**, the imprint head **120**, the fluid dispenser **122**, the radiation source **126**, the thermal radiation source **134**, the field camera **136** and/or the droplet inspection system **138**. The processor **140** may operate based on instructions in a computer readable program stored in a non-transitory computer readable memory **142**. The processor **140** may be or include one or more of a CPU, MPU, GPU, ASIC, FPGA, DSP, and a general purpose computer. The processor **140** may be a purpose built controller or may be a general purpose computing device that is adapted to be a controller. Examples of a non-transitory computer readable memory include but are not limited to RAM, ROM, CD, DVD, Blu-Ray, hard drive, networked attached storage (NAS), an intranet connected non-transitory computer readable storage device, and an internet connected non-transitory computer readable storage device.

(32) Either the imprint head **120**, the substrate positioning stage **106**, or both varies a distance between the mold **110** and the substrate **102** to define a desired space (a bounded physical extent in three dimensions) that is filled with the formable material **124**. For example, the imprint head **120** may apply a force to the template **108** such that mold **110** is in contact with the formable material **124**. After the desired volume is filled with the formable material **124**, the radiation source **126** produces actinic radiation (e.g., UV, 248 nm, 280 nm, 350 nm, 365 nm, 395 nm, 400 nm, 405 nm, 435 nm, etc.) causing formable material **124** to cure, solidify, and/or cross-link; conforming to a shape of the substrate surface **130** and the patterning surface **112**, defining a patterned layer on the substrate **102**. The formable material **124** is cured while the template **108** is in contact with formable material **124** forming the patterned layer on the substrate **102**. Thus, the nanoimprint lithography system **100** uses an imprinting process to form the patterned layer which has recesses and protrusions which are an inverse of the pattern in the patterning surface **112**. In an alternative embodiment, the nanoimprint lithography system **100** uses an imprinting process to form the planar layer with a featureless patterning surface **112**.

(33) The imprinting process may be done repeatedly in a plurality of imprint fields that are spread across the substrate surface **130**. Each of the imprint fields may be the same size as the mesa **110** or just the pattern area of the mesa **110**. The pattern area of the mesa **110** is a region of the patterning

surface **112** which is used to imprint patterns on a substrate **102** which are features of the device or are then used in subsequent processes to form features of the device. The pattern area of the mesa **110** may or may not include mass velocity variation features which are used to prevent extrusions. In an alternative embodiment, the substrate **102** has only one imprint field which is the same size as the substrate **102** or the area of the substrate **102** which is to be patterned with the mesa **110**. In an alternative embodiment, the imprint fields overlap. Some of the imprint fields may be partial imprint fields which intersect with a boundary of the substrate **102**.

(34) The patterned layer may be formed such that it has a residual layer having a residual layer thickness (RLT) that is a minimum thickness of formable material **124** between the substrate surface **130** and the patterning surface **112** in each imprint field. The patterned layer may also include one or more features such as protrusions which extend above the residual layer having a thickness. These protrusions match the recesses **114** in the mesa **110**.

(35) Template/Superstrate

(36) FIG. 2 is an illustration of a template **108** that may be used in an embodiment. The patterning surface **112** may be on a mesa **110** (identified by the dashed box in FIG. 2). The mesa **110** is surrounded by a recessed surface **244** on the front side of the template. Mesa sidewalls **246** connect the recessed surface **244** to patterning surface **112** of the mesa **110**. The mesa sidewalls **246** surround the mesa **110**. In an embodiment in which the mesa is round or has rounded corners, the mesa sidewalls **246** refers to a single mesa sidewall that is a continuous wall without corners.

(37) An alternative plate may be used in another embodiment, referred herein as a superstrate. In the case of the superstrate, the patterning surface **112** is featureless. That is, in an embodiment there is no pattern on the surface **112**. A superstrate with no pattern is used in a planarization process. Thus, when a planarization process is performed, the superstrate is used in place of the template shown in FIG. 1.

(38) Imprinting/Planarizing Process

(39) FIG. 3 is a flowchart of an imprinting process **300** by the nanoimprint lithography system **100** that can be used to form patterns in formable material **124** on one or more imprint fields (also referred to as: pattern areas or shot areas). The imprinting process **300** may be performed repeatedly on a plurality of substrates **102** by the nanoimprint lithography system **100**. The processor **140** may be used to control imprinting process **300**.

(40) In an alternative embodiment, a similar process may be performed to planarize the substrate **102**. In the case of planarizing, substantially the same steps discussed herein with respect to FIG. 3 are performed, except that a patternless superstrate is used in place of the template. Thus, it should be understood that the following description is also applicable to a planarizing method. When using as superstrate, the superstrate may be the same size or larger than the substrate **102**.

(41) The beginning of the imprinting process **300** may include a template mounting step causing a template conveyance mechanism to mount a template **108** onto the template chuck **118**. The imprinting process may also include a substrate mounting step, the processor **140** may cause a substrate conveyance mechanism to mount the substrate **102** onto the substrate chuck **104**. The substrate may have one or more coatings and/or structures. The order in which the template **108** and the substrate **102** are mounted onto the nanoimprint lithography system **100** is not particularly limited, and the template **108** and the substrate **102** may be mounted sequentially or simultaneously.

(42) In a positioning step, the processor **140** may cause one or both of the substrate positioning stage **106** and/or a dispenser positioning stage to move an imprint field *i* (index *i* may be initially set to 1) of the substrate **102** to a fluid dispense position below the fluid dispenser **122**. The substrate **102**, may be divided into *N* imprint fields, wherein each imprint field is identified by an index *i*. In which *N* is a real integer such as 1, 10, 75, etc. $\{N \in \text{custom character.sup.+}\}$. In a dispensing step **S302**, the processor **140** may cause the fluid dispenser **122** to dispense formable material onto an imprint field *i*. In an embodiment, the fluid dispenser **122** dispenses the formable

material **124** as a plurality of droplets. The fluid dispenser **122** may include one nozzle or multiple nozzles. The fluid dispenser **122** may eject formable material **124** from the one or more nozzles simultaneously. The imprint field **i** may be moved relative to the fluid dispenser **122** while the fluid dispenser is ejecting formable material **124**. Thus, the time at which some of the droplets land on the substrate may vary across the imprint field **i**. In an embodiment, during the dispensing step **S302**, the formable material **124** may be dispensed onto a substrate in accordance with a drop pattern. The drop pattern may include information such as one or more of position to deposit drops of formable material, the volume of the drops of formable material, type of formable material, shape parameters of the drops of formable material, etc.

(43) After, the droplets are dispensed, then a contacting step **S304** may be initiated, the processor **140** may cause one or both of the substrate positioning stage **106** and a template positioning stage to bring the patterning surface **112** of the template **108** into contact with the formable material **124** in imprint field **i**.

(44) During a spreading step **S306**, the formable material **124** then spreads out towards the edge of the imprint field **i** and the mesa sidewalls **246**. The edge of the imprint field may be defined by the mesa sidewalls **246**. How the formable material **124** spreads and fills the mesa can be observed via the field camera **136** and may be used to track a progress of a fluid front of formable material.

(45) In a curing step **S308**, the processor **140** may send instructions to the radiation source **126** to send a curing illumination pattern of actinic radiation through the template **108**, the mesa **110** and the patterning surface **112**. The curing illumination pattern provides enough energy to cure (polymerize) the formable material **124** under the patterning surface **112**.

(46) In a separation step **S310**, the processor **140** uses one or more of the substrate chuck **104**, the substrate positioning stage **106**, template chuck **118**, and the imprint head **120** to separate the patterning surface **112** of the template **108** from the cured formable material on the substrate **102**.

(47) If there are additional imprint fields to be imprinted then the process moves back to step **S302**. In an embodiment, additional processing is performed on the substrate **102** in a processing step **S312** so as to create an article of manufacture (e.g., semiconductor device or an optical component). In an embodiment, each imprint field includes a plurality of devices.

(48) The further processing in processing step **S312** may include etching processes to transfer a relief image into the substrate that corresponds to the pattern in the patterned layer or an inverse of that pattern. The further processing in processing step **S312** may also include known steps and processes for article fabrication, including, for example, curing, oxidation, layer formation, deposition, doping, planarization, etching, formable material removal, dicing, bonding, and packaging, and the like. The substrate **102** may be processed to produce a plurality of articles (devices).

(49) Supply and Dispensing System

(50) As noted above, the nanoimprint lithography system **100** includes a supply and dispensing system **200**, alternatively referred herein as a “flow system.” The supply and dispensing system **200** generally includes two or more material management systems and one or more flush systems. FIG. 4 shows a schematic representation of an example embodiment of the supply and dispensing system **200**. As shown in FIG. 4, the example supply and dispensing system **200** includes a first material management system (MMS) **202**, a second MMS **204**, a third MMS **206**, a flush system **208**, a supply manifold **210**, a return manifold **212**, a pressure manifold **214**, and one or more dispensers **122**. The one or more dispensers **122** are alternatively referred herein as “dispenser(s).” Each of the first MMS **202**, the second MMS **204**, the third MMS **206**, and the flush system **208** are in fluid communication with the one or more dispensers **122** via a supply flow path **218a**, **218b**, **218c**, **218d** and return flow path **220a**, **220b**, **220c**, **220d**. Each of the supply flow paths **218a**, **218b**, **218c**, **218d** travel from the corresponding MMS/flush system **202**, **204**, **206**, **208**, through a supply three-way valve **222a**, **222b**, **222c**, **222d**, through the supply manifold **210**, through the pressure manifold **214**, and finally to the one or more dispensers **122**. Each of the return flow paths

220a, 220b, 220c, 220d travels from the one or more dispensers **122**, through the pressure manifold **214**, through the return manifold **212**, through a return three-way valve **224a, 224b, 224c, 224d**, and finally returning to the corresponding MMS/flush system **202, 204, 206, 208**. In FIG. 4, the supply paths have been designated with dashed lines and the return paths have been designated with dotted lines. As also shown in FIG. 4, and as will be discussed in more detail below, because each of the MMS/flush systems **202, 204, 206, 208** has a supply three-way valve **222a, 222b, 222c, 222d** and a return three-way valve **224a, 224b, 224c, 224d** associated with it, it is possible to configure the flow of material exiting one or more of the MMS/flush systems to enter a recycle bypass **226a, 226b, 226c, 226d**. In such a case, the fluid exiting the MMS/flush system enters the supply three-way valve **222a, 222b, 222c, 222d**, where the supply three-way valve **222a, 222b, 222c, 222d** is configured to route the fluid into the recycle bypass line **226a, 226b, 226c, 226d** so that the fluid travels directly to the return three-way valve **224a, 224b, 224c, 224d** without traveling through any of the supply manifold **210**, the pressure manifold **214**, the dispenser(s) **122**, and the return manifold **212**.

(51) The dispenser(s) **122** are the same dispensers discussed above with respect to FIG. 1. The supply manifold **210** is a fitting that receives fluid from multiple separate sources (i.e., from the separate MMS and flush systems) via separate supply tubes and directs the fluid to one or more outlet ports. The supply manifold **210** has a number of inlet ports equal to the total number of MMS and flush systems and one or more outlet ports. After fluid passes through the supply manifold **210** and exits the one or more outlet ports, the exiting fluid is combined into a single outlet conduit leading to the pressure manifold **214**. As shown in FIG. 4, four valves **216a, 216b, 216c, 216d** may be used to allow or prevent fluid from a particular MMS or flush system from entering into the supply manifold **210**. The first valve **216a** (first supply valve) is in communication with the first MMS **202**, the second valve **216b** (second supply valve) is in communication with the second MMS **204**, the third valve **216c** (third supply valve) is in communication with the third MMS **206**, and the fourth valve **216d** (fourth supply valve) is in communication with the flush system **210**. Depending on which of the valves **216a, 216b, 216c, 216d** are open, fluid will be able to flow into the supply manifold **210** and out to the pressure manifold **214**.

(52) The return manifold **212** is the reverse of the supply manifold **210**. That is, the return manifold **212** is a fitting that receives fluid from a single input (i.e., from the pressure manifold **214**) and then directs the fluid to multiple separate outputs (i.e., to the separate MMS and flush systems) via separate return tubes. The return manifold **212** has a number of outlet ports equal to the total number of MMS and flush systems and one or more inlet ports. After fluid coming from the pressure manifold **214** passes into the return manifold **212**, the exiting fluid is directed into one of the outlet ports. As shown in FIG. 4, four valves **227a, 227b, 227c, 227d** may be used to allow or prevent the flow of fluid to a particular one of the MMS or flush systems. The first valve **227a** (first return valve) is in communication with the first MMS **202**, the second valve **227b** (second return valve) is in communication with the second MMS **204**, the third valve **227c** (third return valve) is in communication with the third MMS **206**, and the fourth valve **227d** (fourth return valve) is in communication with the flush system **210**. Depending on which of the valves **216a, 216b, 216c, 216d** are open, fluid will flow into the supply manifold **210** and out to the pressure manifold **214**.

(53) The pressure manifold is has independent supply and return fluid inputs and outputs. The pressure manifold has a supply pressure sensor, supply temperature sensor, return pressure sensor, and return temperature sensor. The sensors provide feedback to the MMS's on how the pressure and temperature should be adjusted (via pumps and heat exchangers) to meet the specified target pressure and target temperature of the formable material at the dispenser. The supply and return pressures are controlled to be close to each other to control the meniscus at the dispenser.

(54) While three different MMS and one flush are shown in FIG. 4, the number of MMS and flush systems is not limited. That is, the method described herein is applicable to a system with as many MMS and flush systems as desired. For example, flow system may include 2 to 10 MMS and 1 to 3

flush systems.

(55) It should be understood that FIG. 4 is generic and shows all possible flow paths to/from the MMS/flush systems, as opposed to particular instances where certain flow paths are active and certain flow paths are inactive. That is, FIG. 4 does not show any particular state/moment of the method described herein. The controller 140 is configured to control all of the controllable elements of the supply and dispensing system 200, i.e., pumps, valves, etc., which changes the flow states. That is, the controller 140 is configured to perform the method 700 described below.

(56) FIG. 5 shows a schematic representation of an example embodiment of a MMS. Each of the MMS 202, 204, 206 have the same components as illustrated in FIG. 5. The only difference between the MMS 202, 204, 206 is that each one could have a different fluid/formable material than the others. That is, while the fluid in every MMS is the formable material described above, the particular formulation may be different depending on the particular fabrication. The formable material for each MMS may be a photocurable composition comprising a photoinitiator and monomers. Exemplar monomers which may be in the photocurable composition include: acrylate monomers; vinyl monomers; styrenic monomers; etc. The formable material may have the composition described in U.S. Pat. App. Pub. No. 2020/0339828, which is hereby expressly incorporated by reference herein. As discussed in U.S. Pat. App. Pub. No. 2020/0339828, the formable material may be a photocurable composition comprising a polymerizable material and a photoinitiator, wherein at least 90 wt % of the polymerizable material may comprise acrylate monomers including an aromatic group. The photocurable composition can have a viscosity of not greater than 10, 15, 20, or 30 mPa-s, the total carbon content of the photocurable composition after curing can be at least 73%, and the Ohnishi number may be not greater than 3.0. At least 90 wt % of the polymerizable material can include monomers containing an aromatic group in their chemical structure. Some non-limiting examples of monomers comprising an aromatic group can be: benzyl acrylate (BA), benzyl methacrylate (BMA), 1-naphthyl methacrylate (1-NMA), bisphenol A dimethacrylate (BPADMA), 1-naphthyl acrylate (1-NA), 2-naphthyl acrylate (2-NA), 9,9-bis[4-(2-acryloyloxy ethoxy) phenyl]fluorine (A-BPEF), 9-fluorene methacrylate (9-FMA), 9-fluorene acrylate (9-FA), o-phenylbenzyl acrylate (o-PBA), bisphenol A diacrylate (BPADA), propenoic acid, 1,1'-[1,1'-binaphthalene]-2,2'-diyl ester (BNDA), styrene, divinyl benzene (DVB). Further details of the composition may be found in U.S. Pat. App. Pub. No. 2020/0339828. Some non-limiting examples of suitable monofunctional (meth)acrylates to be included in the polymerizable material are: isobornyl acrylate; 3,3,5-trimethylcyclohexyl acrylate; dicyclopentenyl acrylate; dicyclopentanyl acrylate; dicyclopentenyl oxyethyl acrylate; benzyl acrylate; naphthyl acrylate; 2-phenylethyl acrylate; 2-phenoxyethyl acrylate; phenyl acrylate; (2-ethyl-2-methyl-1,3-dioxolan-4-yl)methyl acrylate; o-phenyl benzyl acrylate; butyl acrylate; ethyl acrylate; methyl acrylate; n-hexyl acrylate; 2-ethyl hexyl acrylate; 4-tert-butylcyclohexyl acrylate; methoxy polyethylene glycol (350) monoacrylate; 2-methoxyethyl acrylate; lauryl acrylate; stearyl acrylate; 9-fluorene acrylate. Some non-limiting examples of suitable diacrylates to be included in the polymerizable material are: ethylene glycol diacrylate; diethylene glycol diacrylate; triethylene glycol diacrylate; tetraethylene glycol diacrylate; 1,2-propanediol diacrylate; dipropylene glycol diacrylate; tripropylene glycol diacrylate; polypropylene glycol diacrylate; 1,3-propanediol diacrylate; 1,4-butanediol diacrylate; 2-butene-1,4-diacrylate; 1,3-butylene glycol diacrylate; 3-methyl-1,3-butanediol diacrylate; 1,5-pentanediol diacrylate; 3-Methyl-1,5-pentanediol diacrylate; neopentyl glycol diacrylate; tricyclodecane dimethanol diacrylate; 1,6-hexanediol diacrylate; 1,9-nonanediol diacrylate; 1,10-decanediol diacrylate; 1,12-dodecanediol diacrylate; cyclohexane dimethanol diacrylate; bisphenol A diacrylate; ethoxylated bisphenol A diacrylate; m-xylylene diacrylate; 9,9-bis[4-(2-acryloyloxy ethoxy) phenyl]fluorine; 2,2'-diacrylate-1,1'-binaphthalene; dicyclopentanyl diacrylate; 1,2-adamantanediol diacrylate; 2,4-diethylpentane-1,5-diol diacrylate; poly(ethylene glycol) diacrylate; 1,6-hexanediol (EO)2 diacrylate; 1,6-hexanediol (EO)5 diacrylate; and alkoxylated aliphatic diacrylate esters. Some non-limiting examples of suitable multifunctional

acrylates to be included in the polymerizable material are: trimethylolpropane triacrylate; propoxylated trimethylolpropane triacrylate (e.g., propoxylated (3) trimethylolpropane triacrylate, propoxylated (6) trimethylolpropane triacrylate); trimethylolpropane ethoxylate triacrylate (e.g., n~1,3,3,5); di(trimethylolpropane) tetraacrylate; propoxylated glyceryl triacrylate (e.g., propoxylated (3) glyceryl triacrylate); 1,3,5-adamantanetriol triacrylate; tris (2-hydroxy ethyl) isocyanurate triacrylate; pentaerythritol triacrylate; Trisphenol PA triacrylate; pentaerythritol tetracrylate; ethoxylated pentaerythritol tetracrylate; dipentaerythritol pentaacrylate; tripentaerythritol octaacrylate; trimethylolpropane(PO)_n triacrylate (n is 1, 2, 3 . . . trimethylolpropane(EO)_n triacrylate (n is 1, 2, 3 . . .). Examples of the vinyl benzene type of monomers include vinylbenzene (styrene), divinylbenzene (DVB), trivinylbenzene (TVB), 3,3'-divinylbiphenyl, 3,4', 5-trivinylbiphenyl, 3,3',5,5'-tetravinylbiphenyl, 1,2-bis(3-vinylphenyl)ethane, bis(4-vinylphenyl) ether, bis(3-vinylphenyl) ether. Some non-limiting examples of suitable multifunctional monomers to be included in the polymerizable material are: molecules containing both acrylate functional groups and vinyl groups directly connected to aromatic rings. For example, 3-vinyl benzyl acrylate, 2-(4-vinyl)-phenyl, 1,3-propane diacrylate, 3,5-bivinyl benzyl acrylate, and 5-vinyl, 1,3-xylene diacrylate. Some non-limiting examples of maleimides and bismaleimides to be included in the polymerizable material are: N-benzylmaleimide; N-cyclohexylmaleimide; N-phenylmaleimide; and bis(3-ethyl-5-methyl-4-maleimidophenyl)methane. Some non-limiting examples of suitable benzoxazines to be included in the polymerizable material are: 6,6'-Methylenebis[3,4-dihydro-3-phenyl-2H-1,3-benzoxazine; and 3,3'-(Methylenedi-4,1-phenylene)bis[3,4-dihydro-2H-1,3-benzoxazine. As also discussed in U.S. Pat. App. Pub. No. 2020/0339828, the formable material may include additives such as stabilizers, dispersants, solvents, surfactants, inhibitors or any combination thereof. The formable material in each MMS being different from each other MMS for different fabrications means that one or more aspects of the composition may be different, i.e., not identical. For example, the formable materials may be different in the selection of the components (i.e., different or additional monomers, different or additional photoinitiators, different or additional surfactants, and/or different or additional solvents) and/or may be different with respect to the amount of each component (i.e., based on the total mass of the formable material, a different amount of monomer, a different amount of photoinitiator, a different amount of surfactant, and/or a different amount of solvent).

(57) As shown in FIG. 5, each MMS generally includes a plurality of conduits **502**, a plurality of pumps **504**, a plurality of pulsation dampeners **506**, a filter **508**, and a reservoir **510**. The reservoir **510** contains the formable material **124** (fluid, fluid material). The reservoir **510** may be filled with the fluid **124** from a fluid source (not shown) through the conduit **502** connecting the fluid source to the reservoir **510** by action of the appropriate pump **504**. Similarly, the reservoir **510** may be drained of the fluid through the conduit **502** connecting the reservoir **510** to the drain by action of the appropriate pump **504**. The MMS may further include a pressure sensor **512** and level sensor **514**. The fluid in the reservoir **510** is pumped out of the MMS and into the flow system through the conduit **502** heading in the supply direction by action of the appropriate pump **504** in the supply line. After exiting the MMS, the fluid travels through one of various paths described below, depending on the state set by the controller **140**. For example, as described below, the fluid may travel to the dispenser(s) **122** in one state (a dispensing state) or return immediately back without traveling to the dispenser(s) **122** in another state (a bypass recycle state). The fluid similarly returns to the reservoir **510** by action of one of the pumps **504** through the conduit **502** heading in the return direction. During exit and return, the pulsation dampeners **506** dampen pulsation of the fluid that is otherwise introduced by the pumps **504**. The filter **508** filters the fluid each time the fluid exits through the supply line. The filter may be a polyethylene semiconductor-grade filter with 10 nm particle retention size, for example. The particular filter may be optimized based on the particular fluid being circulated. The applicant has also found that the performance of the system **100** being fed by fluid supply and dispensing system **200** is greatly improved if the formable

material is constantly being recirculated through filter even when the formable material is not being used. The filters can continuously remove contaminants and/or larger particles.

(58) FIG. 6 shows a schematic representation of an example embodiment of the flush system **208**. As shown in FIG. 6, the flush system generally includes a plurality of conduits **602**, a plurality of pumps **604**, a plurality of three-way valves **606**, a plurality of filters **608**, an active reservoir **610**, and a waste reservoir **612**. The active reservoir **610** contains a flush fluid **614a** that is to be sent through the flush system via one of the conduits **602** in the supply direction. The flush fluid may be a solvent that is compatible with the formable materials being dispensed and circulated through the system. Compatible means that the flush fluid will dissolve the formable material without reacting with it. For example, the flush fluid may be isopropyl alcohol, acetone, and/or propylene glycol methyl ether acetate (PGMEA). The waste reservoir **612** contains flush fluid **614b** that has already been used and returned to the flush system in the case that flush fluid cannot be further used. The flush fluid in the active reservoir **610** is pumped out of the flush system toward the flow system by action of one of the pumps **604** through the conduit **602** heading in the supply direction. After exiting the flush system, the flush fluid travels through one of various paths described below, depending on the state set by the controller **140**. For example, as described below, the flush fluid may travel to the dispenser(s) **122** in one state (a dispensing state) or return immediately back without traveling to the dispenser(s) **122** in another state (a bypass recycle state). The flush fluid similarly returns to either the active reservoir **610** or the waste reservoir **612** by action of one of the pumps **604** through the conduit **602** heading in the return direction. In an embodiment, the pump **604** that pulls the fluid through the return line may be a venturi that is used to produce suction. The filter **608** may be the same filters as described above and serves the function of filtering the flush fluid as it exits the active reservoir **610** in the direction of the supply line. The flush system **208** further includes a drying gas source **616** so that a supply of drying gas can be controlled to flow through the conduits **602** in the supply direction. By controlling the three-way valves **606** and the pumps **604**, the flush system **608** can be controlled to supply flush fluid **614a** from the active reservoir **610** through the supply line, to provide drying gas from the drying gas source **616** through the supply line, and to either have the flush fluid returning through the return line to flow back to the active reservoir **610** or to flow into the waste reservoir **612**. The flush system **208** may include a valve assembly **618** that includes one or more valves, two input ports, and two output ports. The valve assembly **618** can be switched from between two states. In the first state the valve assembly **618** provides: a first fluid connection between the return pump **604** and the return line; and a second fluid connection between the supply line three-way valve **606** and the supply line. In the second state the valve assembly **618** provides: a third fluid connection between the return pump **604** and the supply line; and a fourth fluid connection between the supply line three-way valve **606** and the return line.

(59) It should be noted that FIGS. 5 and 6 are generic and shows all possible flow paths to/from the MMS/flush systems, as opposed to particular instances where certain flow paths are active and certain flow paths are inactive. That is, FIGS. 5 and 6 do not show any particular state/moment of the method described herein. The controller **140** is configured to control all of the controllable elements of MMS and flush systems, i.e., pumps, valves, etc., which changes the flow states.

(60) Method of Switching Dispensed Fluid

(61) Using controller **120** to control the various pumps, valves, etc. of the supply and dispensing system **200**, a method **700** of switching a fluid supplied to dispenser(s) with a second liquid can be performed. FIG. 7 is a flow chart of the method **700**.

(62) As shown in FIG. 7, the first step in the method **700** is step **S702**, where a first fluid is stopped from flowing into the common portion **228** of the supply and dispensing system **200** (flow system). The common portion **228** is the portion of the supply and dispensing system **200** that includes the supply manifold **210**, the one or more dispensers **122**, the return manifold **212**, a supply portion **230** providing fluid communication between the supply manifold **210** and the one or more dispensers **122**, and a return portion **232** providing fluid communication between the one or more

dispensers **122** and the return manifold **212**. The common portion **228** further includes the pressure manifold **214**. The supply portion **230** includes a conduit **234a** connecting the supply manifold **210** to the pressure manifold **214** and a conduit **234b** connecting the pressure manifold **214** to the one or more dispensers **122**. The return portion **232** includes a conduit **236a** connecting the one or more dispensers **122** to the pressure manifold **214** and a conduit **236b** connecting the pressure manifold **214** to the return manifold **212**. The common portion **228** does not include the conduits connecting the MMS and flush systems to the supply manifold, does not include the conduits connecting the return manifold to the MMS and flush systems, does not include the recycle bypass, and does not include any of the three-way valves or any of the valves at the supply and return manifold.

Similarly, the common portion **228** does not include any of the components of the MMS and flush systems. In other words, the common portion **228** does not include any of the components upstream of the supply manifold nor any of the components downstream of the return manifold.

(63) FIG. **8** shows the supply and dispensing system **200** at a moment just before performing step **S702**. That is, the start of the method **700** presumes a preexisting active state of the supply and dispensing system **200** where one of the MMS is actively providing fluid (formable material) to the dispenser and back to the MMS. In the example embodiment shown in FIG. **8**, the first MMS **202** is the active MMS, i.e., the MMS providing the fluid that is actively being dispensed from the dispenser(s) **122**. However, any one of the MMS may be active prior to step **S702**. As noted above, the details of the first MMS **202** is shown in FIG. **5**. The fluid material **124** begins in the reservoir **510**. The fluid material **124** is pumped from the reservoir **510** using the pump **502** in the supply line, through the filter **508**, through the pulsation dampener **506** in the supply line, and then out of first MMS **202**. After exiting the first MMS **202**, the fluid material enters the portion of the flow system illustrated in FIG. **8**.

(64) In FIG. **8**, the solid lines indicate an “active flow” state and the dashed lines indicate a “no flow” state. As shown in FIG. **8**, prior to step **S702**, fluid is flowing from the first MMS **202** into the supply three-way valve **222a**, where the supply three-way valve **222a** is directing the flow through the supply flow path **218a** while blocking the flow through the recycle bypass **226a**. Thus, the fluid material coming from the first MMS **202** continues to flow to the supply manifold **210** via the open supply valve **216a** (the other supply valves being in a closed state), out of the supply manifold **210**, into the pressure manifold **214** via the conduit **234a**, and into the one or more dispensers **122** via the conduit **234b**. Next, the fluid material that is not dispensed travels back through the pressure manifold **214** via the conduit **236a**, to the return manifold **212** via the conduit **236b**, out the return manifold **212** via the return valve **227a** (the other return valves being in a closed state), and through the return three-way valve **224a**. The return three-way valve **224a** directs the returning fluid back into the first MMS **202**. Thus, in state shown in FIG. **8**, the fluid material coming from the first MMS **202** travels from the first MMS **202** to the one or more dispensers **122** and returns back to the first MMS **202**. Once returning to the first MMS **202**, as shown in FIG. **5**, the fluid material is pumped by the pump **502** in the return line, through the pulsation dampener **506** in the return line, and back into the reservoir **510**. In this flow state, the pumps **502** are configured to supply a positive pressure on the fluid entering the dispenser(s) **122** and negative pressure on the fluid exiting the dispenser(s) **122**. The differential between the positive and negative pressure can be used to control the meniscus of fluid in the nozzles of the dispenser(s) **122**.

(65) As also shown in FIG. **8**, in the state in which only the fluid material of the first MMS **202** is being supplied, all of the other systems are in a recycle bypass state. In the recycle state, the fluid material is flowing from each of the other MMS and flush systems and be rerouted back to the respective systems from which it came. In particular, for example, in the case of the second MMS **204**, fluid material is exiting the second MMS **204** and entering the supply three-way valve **222b**. In the recycle bypass state, the supply three-way valve **222b** blocks the flow of the fluid material from entering into the supply flow path **218b**. Instead, the three-way valve **222b** causes the fluid

material to move through the recycle bypass **226b** and into the return three-way valve **224b**. In the recycle bypass state, the return three-way valve **224b** further directs the fluid material flowing from the supply three-way valve **222b** back to the second MMS **204**. That is, in recycle bypass state, instead of the fluid material traveling through the supply path **218b**, to the one or more dispensers **122**, and then returning through the return flow path **220b**, the fluid material bypasses all of those components and returns directly to the second MMS **204**. Turning to FIG. 5, regardless of whether the fluid returns to the second MMS **202** via the recycle bypass path or returns after the dispensing route, the treatment of the fluid material returning to the second MMS **204** is the same. That is, the fluid is similarly pumped into the reservoir **510** and back out of the system through the filter **508**. Accordingly, in the bypass recycle state, the fluid material is continuously cycling into and out of the MMS and passing through the filter **508** on every cycle. The filtering of the fluid material in every cycle provides the advantage of minimizing contamination in the fluid. Thus, continuously recycling the fluid minimizes, avoids, and/or reduces contamination in the fluid.

(66) The description of the fluid traveling out of and back into the second MMS **204** above is essentially the same as the fluid that travels out of and back into the third MMS **206** and the flush system **208** shown in FIG. 8. That is, as shown in FIG. 8, the third MMS **206** is similarly in the recycle bypass state where the fluid material exits the third MMS **206**, and is routed directly back to the third MMS **206** via the three-way valves **222c**, **224c** and the recycle bypass **226c**. Likewise, fluid material is also exiting the flush system **208** and being routed directly back to the flush system **208** via the three-way valves **222d**, **224d** and the recycle bypass **226d**. In the case of the flush system **208**, the returning fluid may or may not be reused. In the case that the flush fluid is reused, the three-way valve **606** in the return line is set to direct the returning flush fluid to travel back into the active reservoir **610** while blocking the returning flush fluid from entering the waste reservoir **612**. The flush fluid **614a** in the active reservoir **610** is then pumped back out through the filter **608** through the supply line. In contrast, when the returning flush fluid is not being reused, the three-way valve **606** in the return line is set so that the returning flush fluid is directed to the waste reservoir **612**. In this case, the supply of flush fluid continues from the active reservoir **610**, but none of the returning flush fluid comes back into the active reservoir **610**. In either configuration, the three-way valve **606** in the supply line is set so that the drying gas does not enter the supply line and only the flush fluid **614a** enters the supply line. In an alternative embodiment, the flushing fluid is never recirculated when not being used for flushing. In an alternative embodiment, flushing fluid is sometimes recirculated when not being used for flushing, when filtering of the flushing fluid is desired.

(67) FIG. 9 shows the supply and dispensing system **200** at a moment after performing step S702. That is, FIG. 9 shows the moment that the first fluid material (in this case the fluid material coming from the first MMS **202**) is stopped from flowing into the common portion **228** of the flow system. As shown in FIG. 9, as indicated by the dashed “no flow” lines, fluid material is not flowing into the common portion **228**. All of the fluid that was entering the system from the first MMS **202** is returned to the first MMS **202** and enters the reservoir **510**. Further, the flow of fluid material is stopped from flowing out of the first MMS **202**. That is, at this moment, the pumps **502** are turned off so fluid is no longer flowing out of the reservoir **510** and into the supply line. In an embodiment, the supply pump of the MMS **202** is turned off, then the return pump of the MMS **202** is turned off a few moments later. At the same time, as also shown in FIG. 9, the second MMS **204** and the third MMS **206** are held in the recycle bypass state, i.e., the same state as in FIG. 8. That is, the fluid material coming out of the second MMS **204** and the third MMS **206** continues to follow the recycle bypasses **226b**, **226c**, respectively, without being shut off, even as the flow of the fluid material from the first MMS **202** is terminated.

(68) Because the flush system **208** is about to be used to flush out the common portion **228**, in the illustrated example embodiment, the flow of flush fluid from the flush system **208** has also been terminated at the moment shown in FIG. 9. That is, while the fluid material of the flush system **208**

had been in the recycle bypass state when the fluid material of the first MMS 202 was actively flowing to the dispenser(s) 122, the recycle bypass of the flush fluid from the flush system may similarly be terminated. Thus, as shown in FIG. 9 via the dashed “no flow” lines, there is no fluid material exiting the flush system 208 at the moment shown in FIG. 9. That is, at this moment the pumps 604 are turned off so that no flush fluid is exiting or entering the flush system 208.

(69) After completing step S702, the method may proceed to step S704, where the flush fluid is caused to flow through the common portion 228 of the supply and dispensing system 200. FIG. 10 shows the supply and dispensing system 200 at the moment of performing step S704. As shown in FIG. 10, represented by the solid “active flow” lines, the flush fluid is departing the flush system 208 and traveling through the supply three-way valve 222d toward the supply manifold 210. The supply three-way valve 222d is controlled to have a configuration where the flush fluid is blocked from traveling through the recycle bypass 226d and instead enters into the supply manifold 210 via the now-opened fourth valve 216d. The flush fluid travels through the supply portion 230, i.e., to the pressure manifold 214 via the conduit 234a and to the dispenser(s) 122 via the conduit 234b. Some of the flush fluid is dispensed out of the dispenser(s) 122. The flush fluid that is not dispensed travels through the return portion 232, i.e., to the pressure manifold 214 via the conduit 236a and then to the return manifold via the conduit 236b. The flush fluid exits the return manifold 212 via the return valve 227d (the other return valves being in a closed state), and through the return three-way valve 224d. The return three-way valve 224d directs the returning flush fluid back into the flush system 208. Thus, in state shown in FIG. 10, the flush fluid coming from the flush system 208 travels from the flush system 208 to the one or more dispenser(s) 122 and returns back to the flush system 122. While the flushing fluid is being supplied the dispenser(s) 122 may be in a state (for example a jetting state, cleaning state, flushing state, etc.) that allows the formable material to be removed from the nozzles of the dispenser(s) 122. The dispenser(s) may also bin in a maintenance position which ensures that any fluid that does exit the dispenser(s) does not impact the performance of the system 100.

(70) During the state shown in FIG. 10, the pump 604 in the supply line is actively pumping the flush fluid 614a from the active reservoir 610, through the three-way valve 606 in the supply line, and out the supply line, where the three-way valve 606 in the supply line is set so that only the flush fluid passes through and not the drying gas. At the same time, the pump 604 in the return line is actively pumping the non-dispensed flush fluid back into the return line and either back into the active reservoir 610 or into the waste reservoir 612 depending on the setting of the three-way valve 606 in the return line (i.e., depending on whether the flush fluid is to be recycle or is to be sent to waste).

(71) FIG. 10 also illustrates a moment where the step S720 of beginning to continuously circulate the first liquid to and from the reservoir 510 has been performed. That is, in one example embodiment, the step 720 of beginning to continuously circulate the first liquid to and from the reservoir 510 can be performed immediately after the step S702. Once the circulating has begun in step S720, the circulation is maintained throughout all of the remaining steps. However, step S720 need not be performed immediately after step S702. The step S720 of beginning the continuous circulation of the first fluid can be performed at any point in the method all the way up to and including the below-discussed step S710 of causing the second liquid to flow through the common portion 228 of the flow system. In other words, the step S720 of beginning the recycle flow of the first liquid can be performed at any point after step S702 and prior to or during step S720.

(72) Next, the method may proceed to step S706, where the flush fluid is stopped from flowing into the common portion 228 of the flow system. This can be achieved by stopping the pumping of the flush fluid from the active reservoir 610. Specifically, the pump 604 in the supply line of the flush system 208 may stop pumping the flush fluid 614a through the supply line.

(73) After stopping the flow of flush fluid to the common portion 228 of the flow system 200, the method may proceed to step S708 where the flush fluid is removed from the common portion 228

of the flow system. Step **S708** may include using the pump **604** in the return line of the flush system **208** to continue to pump the remaining flush fluid in the flow system back into the either the active reservoir **610** or the waste reservoir **612**. Step **S708** may further include a subsequent step of flowing a drying gas through the common portion **228** to remove any remaining flush fluid. The drying gas serves the function of removing any flush liquid that may still be in the lines. The drying gas may be any inert gas such as clean dry air, nitrogen, argon, neon, and/or helium, for example. Preferably, the drying gas is clean dry air and/or nitrogen. As noted above, the flush system **208** includes a drying gas source **616** which can enter the supply line of the flush system **208** via the—three way valve **606** in the supply line. The three-way valve **606** in the supply line of the flush system **208** is actuated so that the drying gas enters the supply line. The drying gas may then flow through the same “active lines” shown in FIG. **10**. That is, one embodiment, the flow of the drying gas follows the same travel path as the flush fluid described above, including flowing through the common portion **228**. In such a case where the drying gas follows the exact same flow path as the flush fluid, the drying gas will ultimately be expelled into the waste reservoir **612** or the active reservoir **610**. In another embodiment, an additional valve may be included in the flow path just downstream of the return manifold **212** so that the drying gas can be vented prior to returning to the flush system **208**. Because the nozzles of the dispenser(s) are open to air, in order for the flushing fluid and/or drying gas to be removed from the supply line, step **S708** may include a step of switching the valve assembly **618** from the first state to the second state so that the flushing fluid and/or drying gas remaining in the supply line is also removed.

(74) After completing step **S708**, the method may proceed to step **S710**, where the second fluid of the second MMS **204** is caused to flow through common portion **228** of flow system **200**. FIG. **11** shows the flow system **200** at a moment just prior to step **S710**. As part of preparing the flow of the second fluid to flow from the second MMS **204** into the common portion **228** of the flow system **200**, an intermediate step may be performed where the flush system **208** is completely stopped from sending or returning any fluid (flush or drying gas). As shown in FIG. **11**, represented by the dashed “no flow” lines, the flush fluid and gas is no longer departing the flush system **208** at all. That is, neither the flush fluid nor the drying gas is being supplied by the flush system **208** and is therefore neither flowing to/from the dispenser(s) **122** nor traveling through the recycle bypass **226d**. Similarly, as also shown in FIG. **11** via the dashed “no flow” lines, as part of the intermediate preparation step, the second fluid from the second MMS **204** may also be stopped from flowing from the second MMS **204**. That is, the second fluid is not being supplied by the second MMS **204** and is therefore neither flowing to/from the dispenser(s) **122** nor traveling through the recycle bypass **226b**. Regarding the first MMS **202**, however, as noted above, because the step **S720** of continuously recycling the first fluid from the first MMS **202** via the recycle bypass **226a** is maintained through the method, FIG. **11** shows “active flow” solid lines exiting the first MMS **202** and returning back to the first MMS **202** via the recycle bypass **226a**. Similarly, because the illustrated example embodiment is showing the process of switching to the second fluid of the second MMS **204**, the third fluid from the third MMS **206** is also shown traveling out of the third MMS **206** and returning directly back to the third MMS **206** via the recycle bypass **226c**.

(75) Turning to FIG. **12**, after achieving the state shown in FIG. **11**, where the first MMS **202** and the third MMS are in a bypass recycle state, while the second MMS **204** and the flush system **208** are in a stopped state, the method may then proceed to performing step **S710** where the second fluid of the second MMS **204** is caused to flow through common portion **228** of flow system **200**. FIG. **12** shows a state where the second fluid of the second MMS **204** is flowing through the common portion **228**, to the dispenser(s) **122** and eventually back to the second MMS **204**. In particular, as shown in FIG. **12**, fluid is flowing from the second MMS **204** into the supply three-way valve **222b**, where the supply three-way valve **222b** is directing the flow through the supply flow path **218b** while blocking the flow through the recycle bypass **226b**. Thus, the fluid material coming from the second MMS **204** continues to flow to the supply manifold **210** via the open

supply valve **216b** (the other supply valves being in a closed state), out of the supply manifold **210**, into the pressure manifold **214** via the conduit **234a**, and into the one or more dispenser(s) **122** via the conduit **234b**. Next, the fluid material that is not dispensed travels back through the pressure manifold **214** via the conduit **236a**, to the return manifold **212** via the conduit **236b**, out the return manifold **212** via the return valve **227b** (the other return valves being in a closed state), and through the return three-way valve **224b**. The return three-way valve **224b** directs the returning fluid back into the second MMS **204**. Thus, as noted above, in the state shown in FIG. **12**, the fluid material coming from the second MMS **204** travels from the second MMS **204** to the one or more dispensers **122** and returns back to the second MMS **204**. Once returning to the second MMS **204**, the same path is traveled as discussed above with respect to the first MMS **202**. That is, as shown in FIG. **5**, the second fluid material travels through the pulsation dampener **506** of the return line and back into the reservoir **510**. At the same time, the fluid in reservoir **510** is flowing out the supply line passing through the filter **508** and the pulsation dampener **506** of the supply line.

(76) As also shown in FIG. **12**, in the state in which only the fluid material of the second MMS **204** is being supplied, all of the other systems are in a recycle bypass state, as discussed above. Thus, upon reaching the state shown in FIG. **12** by performing the method **700**, the system **200** has switched from supplying a first fluid from a first MMS **202** to the dispenser(s) to supplying a second fluid from a second MMS **204** without having any of the previous material remaining in the common portion **228**. Comparing FIG. **8** to FIG. **12** shows the initial flow state compared to the final switched flow state, i.e., the state prior to performing the method **700** and the state after completing the method **700**. As seen by comparing these figures, the flow system **200** is similar in each state in that in each state only one fluid is being provided to the dispenser(s) **122** while all the other fluids are in a recycle state. The only difference is that in FIG. **8** the first fluid from the first MMS **202** is being supplied, while in FIG. **12** the second fluid from the second MMS **204** is being supplied. Dispenser(s) **122** are now being supplied with fluid from the second MMS **204**. The system **100** may then start using the fluid from the second MMS **204** during the dispensing step **S302** of the process **300**. Prior to the Process **300** being implemented, the dispenser(s) may go through a testing procedure to ensure all of the nozzles of the dispenser(s) are working correctly.

(77) The method described herein can also be applied to switch from any one MMS to any other MMS. That is, while the illustrated example embodiment starts with the first MMS **202** and switches to the second MMS **204**, the same method can be applied to switch from the first MMS **202** to the third MMS **206**, the second MMS **204** to the third MMS **206**, the third MMS **206** to the first MMS **202**, the third MMS **206** to the second MMS **204**, and every other permutation. Furthermore, while only three MMS are illustrated in the example embodiment, any number of MMS can be used in a single system, where additional lines and additional valves are added to the supply and return manifolds.

(78) As noted above, the step **S720** can be performed at any moment during the process after step **702** all the way up to and including at the same as step **S710**. That is, the step of continuously circulating the first liquid (i.e., recycle bypass), can begin as early as just after stopping the flow of the first fluid to the common portion of the flow system and begin as late as the moment when the fluid from another MMS enters the common portion of the flow system. Once the continuous recycle bypass circulation begins, it may be maintained throughout the entire time that any other fluid is passing to/from the dispenser(s). In other words, once the recycle bypass starts for the fluid previously being dispensed, it may stay in the recycle bypass state indefinitely until it is time to return to dispensing that fluid again. For example, a fluid from one of the MMS may be dispensed over the course of several hours to several weeks where the other fluids in the other MMS are continuously recycled during this entire time.

(79) Further modifications and alternative embodiments of various aspects will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. It is to be understood that the forms shown and described herein are to be taken as

examples of embodiments. Elements and materials may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of this description.

Claims

1. A method of switching a first fluid supplied to a dispenser with a second fluid, the method comprising: stopping the first fluid from flowing into a common portion of a flow system, the common portion including: a supply manifold; the dispenser; a return manifold; a supply path providing fluid communication between the supply manifold and the dispenser; and a return path providing fluid communication between the dispenser and the return manifold, causing a flush fluid to flow through the common portion of the flow system; stopping the flush fluid from flowing into the common portion of the flow system; removing the flush fluid from the common portion of the flow system; causing the second fluid to flow through the common portion of the flow system; and continuously circulating the first fluid to and from a reservoir without the first fluid entering the common portion of the flow system, wherein the continuously circulating of the first fluid begins a) after stopping the first fluid from flowing into the common portion of the flow system and b) prior to or during the second fluid flowing through the common portion of the flow system.
2. The method of claim 1, further comprising maintaining the continuously circulating of the first fluid as long as the second fluid flows through the common portion of the flow system.
3. The method of claim 1, wherein the continuously circulating of the first fluid begins simultaneously with causing the flush fluid to flow through the common portion of the flow system.
4. The method of claim 1, wherein the continuously circulating of the first fluid begins simultaneously with causing the second fluid to flow through the common portion of the flow system.
5. The method of claim 1, wherein both the first fluid and second fluid include a polymerizable material.
6. The method of claim 1, wherein the first fluid is different from the second fluid.
7. The method of claim 1, further comprising continuously circulating a third fluid to and from a reservoir without the third fluid entering the common portion of the flow system.
8. The method of claim 1, wherein the removing of the flush fluid further comprises causing a drying gas to flow into the common portion of the flow system.
9. The method of claim 1, wherein the common portion of the flow system further comprises a pressure manifold.
10. The method of claim 1, further comprising, after removing the flush fluid from the common portion of the flow system, continuously circulating the flush fluid to and from a flush reservoir without the flush fluid entering the common portion of the flow system.
11. The method of claim 1, further comprising dispensing the first fluid from the dispenser prior to stopping the first fluid from flowing into the common portion of the flow system.
12. The method of claim 7, wherein the first fluid is different from the second and third fluid; and wherein the second fluid is different from the third fluid.
13. The method of claim 7, wherein the continuously circulating of the third fluid begins prior to the stopping of the first fluid and is maintained through the causing of the second fluid to flow through the common portion of the flow system.
14. The method of claim 8, wherein the drying gas is nitrogen and/or clean dry air.
15. The method of claim 9, wherein the supply path includes supply conduits connecting the supply manifold to the pressure manifold and connecting the pressure manifold to the dispenser.
16. The method of claim 11, further comprising dispensing the second fluid from the dispenser

after causing the second fluid to flow through the common portion of the flow system.

17. The method of claim 15, wherein the return path includes return conduits connecting the dispenser to the pressure manifold and connecting the pressure manifold to the return manifold.

18. A method of making an article comprising: stopping a first fluid from flowing into a common portion of a flow system, the common portion including: a supply manifold; a dispenser; a return manifold; a supply path providing fluid communication between the supply manifold and the dispenser; and a return path providing fluid communication between the dispenser and the return manifold, causing a flush fluid to flow through the common portion of the flow system; stopping the flush fluid from flowing into the common portion of the flow system; removing the flush fluid from the common portion of the flow system; causing a second fluid to flow through the common portion of the flow system; continuously circulating the first fluid to and from a reservoir without the first fluid entering the common portion of the flow system, wherein the continuously circulating of the first fluid begins a) after stopping the first fluid from flowing into the common portion of the flow system and b) prior to or during the second fluid flowing through the common portion of the flow system; dispensing the second fluid onto a substrate using the dispenser; forming a pattern or a layer of the second fluid dispensed on the substrate; and processing the formed pattern or layer to make the article.
